

Application of NPP-VIIRS Nighttime Light Imagery for GDP Spatialization and Economic Differences in Vietnam

Kien Le

Gross domestic product (GDP) is a fundamental economic indicator within a country or region. More detailed GDP estimates can provide researchers and policymakers with a clearer understanding of regional economic disparities and development patterns. However, conventional socioeconomic statistics used to measure GDP often suffer from inconsistent statistical scales and limited spatial resolution, making it difficult to capture regional economic disparities at a fine spatial scale. Therefore, many studies have proposed various methods for constructing maps and data on regional GDP with respect to real world coordinates. Researchers have shown that nighttime light (NTL) data are a useful economic indicator and have adopted and developed methods for spatializing GDP using NTL data, including both statistical and machine learning approaches. However, applications of NTL data for GDP spatialization in Vietnam remain limited, motivating this study to spatialize regional GDP in Vietnam using NTL data.

Remote sensing imagery offers clear advantages over conventional socioeconomic statistics for describing the spatial GDP distribution. Before Visible Infrared Imaging Radiometer Suite (VIIRS) data, carried by the Suomi National Polar-Orbiting Partnership (NPP), became available in 2013, researchers primarily relied on NTL data from the Defense Meteorological Satellite Program-Operational Linescan System (DMSP-OLS). These studies found a strong relationship between NTL intensity and regional GDP. For example, Elvidge et al. (1997) found a strong correlation between the area lit and GDP across 21 counties in North America using DMSP-OLS

NTL data. Similar studies conducted on the data from the United States, Europe, and China have also reported strong relationships between NTL and GDP (Doll et al., 2006; Zhao et al., 2011; Forbes, 2013).

Following the introduction of NPP-VIIRS data in 2013, research increasingly shifted toward using this newer dataset (Li et al., 2013; Zhao et al., 2017). NPP-VIIRS provides higher spatial resolution (approximately 500 m) than DMSP-OLS data (1 km), while its wider radiometric detection range reduces saturation and its onboard radiometric calibration improves data quality (Elvidge et al., 2013). However, NPP-VIIRS imagery is a preliminary product affected by various sources of light, including cities, transportation corridors, gas flares, biomass burning, background noise, and reflections from bright surfaces such as snow covered mountains and playa lake beds (Elvidge et al., 2013). To address these issues, Zhao et al. (2017) proposed a method for correcting VIIRS nighttime light data (see figure 1). Building upon the procedure developed by Zhao et al. (2017), Xu et al. (2024) estimated regional GDP for all provinces and prefectures in China while incorporating land cover data. Their results also highlighted the effectiveness of Geographically Weighted Regression (GWR) in capturing economic development using NTL data. The GWR model explained up to 93% of the variation in economic development, compared with 82% explained by the Ordinary Least Squares (OLS) model (Xu et al., 2024). Liu et al. (2025) used NPP-VIIRS nighttime light and land use data to develop a 1 km gridded GDP dataset for China using machine learning methods, achieving an R-squared of 0.78 and providing a more detailed representation of regional economic disparities.

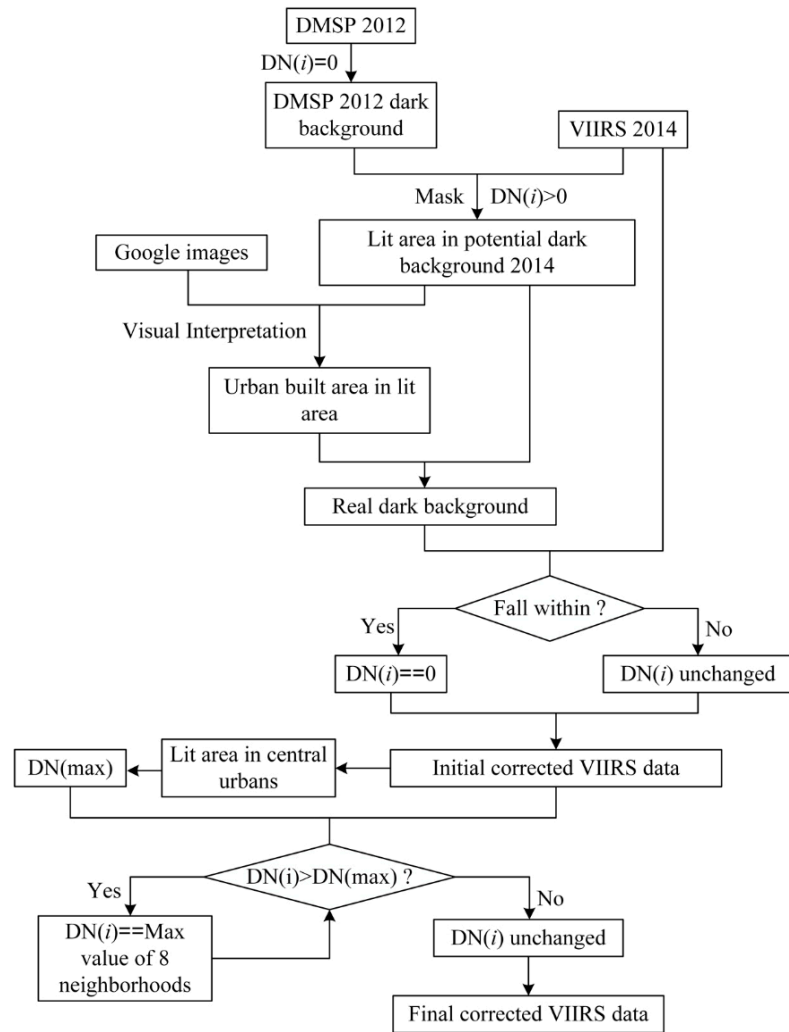


Figure 1: Processing flow chart for lighting correction of VIIRS nighttime light data (Zhao et al., 2017).

For Vietnam, several studies have also used NTL data to analyze and predict economic indicators. Since official GDP data for Vietnam have only been available from 2010 onward, Suleiman et al. (2025) estimated Vietnam's subnational GDP from 1992 to 2010 using NTL data, demonstrating that remote sensing data can serve as a reliable predictor of Vietnam subnational GDP. In addition to that, Zhang (2025) demonstrated that NTL data can effectively capture spatial economic disparities patterns in Vietnam, particularly by identifying clusters of

economically developed and underdeveloped regions. These findings suggest that the relationship between NTL data and economic activity observed in other regions also holds in Vietnam, providing a strong basis for applying NTL methods for GDP spatialization. However, to the best of my knowledge, no existing study has applied NTL data to spatialize Vietnam's GDP at the regional level, although similar applications have been conducted at the global level (Wang et al., 2019).

Therefore, this study aims to address this gap by adapting and extending the methods used in previous research to spatialize regional GDP in Vietnam using NPP-VIIRS NTL data. In addition, this research will develop an automated and reproducible analytical workflow that takes NPP-VIIRS imagery and regional GDP data as inputs and produces spatial GDP estimates.

References

- Doll, C. N. H., Muller, J.-P., & Morley, J. G. (2006). Mapping regional economic activity from night-time light satellite imagery. *Ecological Economics*, 57(1), 75-92.
<https://doi.org/10.1016/j.ecolecon.2005.03.007>
- Elvidge, C. D., Baugh, K. E., Kihn, E. A., Kroehl, H. W., Davis, E. R., & Davis, C. W. (1997). Relation between satellite observed visible-near infrared emissions, population, economic activity and electric power consumption. *International Journal of Remote Sensing*, 18(6), 1373-1379. <https://doi.org/10.1080/014311697218485>
- Elvidge, C. D., Baugh, K. E., Zhizhin, M., & Hsu, F.-C. (2013). Why VIIRS data are superior to DMSP for mapping nighttime lights. *Proceedings of the Asia-Pacific advanced network*, 35(0), 62. <http://dx.doi.org/10.7125/APAN.35.7>

- Forbes, D. J. (2013). Multi-scale analysis of the relationship between economic statistics and DMSP-OLS night light images. *GIScience & remote sensing*, 50(5), 483-499.
<https://doi.org/10.1080/15481603.2013.823732>
- Li, X., Xu, H., Chen, X., & Li, C. (2013). Potential of NPP-VIIRS nighttime light imagery for modeling the regional economy of China. *Remote Sensing*, 5(6), 3057-3081.
<https://doi.org/10.3390/rs5063057>
- Liu, S., Liu, W., Zhou, Y., Wang, S., Wang, F., & Wang, Z. (2025). Mapping gridded GDP distribution of China based on remote sensing data and machine learning methods. *Remote Sensing*, 17(10), 1709. <https://doi.org/10.3390/rs17101709>
- Wang, X., Sutton, P. C., & Qi, B. (2019). Global mapping of GDP at 1 km² using VIIRS nighttime satellite imagery. *ISPRS international journal of geo-information*, 8(12), 580.
<https://doi.org/10.3390/ijgi8120580>
- Xu, N., Zhang, S., & Jiang, S. (2024). Estimating GDP by fusing nighttime light and land cover data. *Canadian Journal of Remote Sensing*, 50(1), 2377641.
<https://doi.org/10.1080/07038992.2024.2377641>
- Zhang, C. (2025). Identifying spatial economic disparities in Vietnam: evidence from nighttime light data. *GeoJournal*, 91(1), 5. <https://doi.org/10.1007/s10708-025-11556-6>
- Zhao, M., Cheng, W., Zhou, C., Li, M., Wang, N., & Liu, Q. (2017). GDP spatialization and economic differences in South China based on NPP-VIIRS nighttime light imagery. *Remote Sensing*, 9(7), 673. <https://doi.org/10.3390/rs9070673>
- Zhao, N., Currit, N., & Samson, E. (2011). Net primary production and gross domestic product in China derived from satellite imagery. *Ecological Economics*, 70(5), 921-928.
<https://doi.org/10.1016/j.ecolecon.2010.12.023>